Customer Shopping Behavior — Analytics Report

Prepared using Python, SQL and Power BI

Executive Summary

This report walks through an end-to-end analysis of a retail transaction dataset covering 3,900 customer purchases. Raw data was cleaned and enriched in Python, business questions were answered directly against the data using SQL, and the results were brought together in an interactive Power BI dashboard. The sections below cover the dataset itself, the cleaning and feature-engineering steps, ten SQL-driven business questions with their results, the dashboard, and a short set of recommendations based on what the numbers show.

3,900	18 → 19	10	1
Transactions Analyzed	Columns Before / After Cleaning	Business Questions Answered in SQL	Power BI Dashboard

1. Dataset at a Glance

The dataset represents individual retail transactions and includes customer demographics, what was purchased, and how the customer shops. Before cleaning, it looked like this:

- Records: 3,900 individual purchases
- Fields: 18 original columns
- Field groups:
  - Who bought it — Age, Gender, Location, Subscription Status
  - What was bought — Item Purchased, Category, Purchase Amount, Season, Size, Color
  - How they shop — Discount Applied, Promo Code Used, Previous Purchases, Purchase Frequency, Review Rating, Shipping Type
- Data quality issue found: 37 missing values in Review Rating, addressed during cleaning.

2. Preparing the Data in Python

Before any analysis could happen, the raw file needed cleaning. This stage was done in Python with pandas, and the cleaned output became the source for both the SQL analysis and the Power BI dashboard.

- **Load:** Read the raw CSV into a DataFrame with pandas.
- **Inspect:** Ran df.info() and df.describe() to understand structure, data types, and value ranges before touching anything.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discou Appli
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	N
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	N
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	N
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	N
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	N
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	N
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	N

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- **Fix missing ratings:** The 37 missing Review Rating values were filled with the median rating for that product's Category, rather than one dataset-wide median, so the imputed scores stayed realistic per product type.
- **Standardize columns:** All column headers were converted to lowercase snake_case for consistency across Python, SQL and Power BI.
- **Engineer features:** Two new columns were added to support deeper analysis:
  - age_group — customers binned into four age ranges
  - purchase_frequency_days — the text frequency values (weekly, monthly, etc.) converted into a numeric day count
- **Remove redundancy:** discount_applied and promo_code_used turned out to hold identical information in every row, so the duplicate promo_code_used column was removed.

- **Hand off to SQL:** The cleaned DataFrame was loaded into a PostgreSQL database so it could be queried directly with SQL.

  Python codes:

```
import pandas as pd
df = pd.read_csv("/content/customer_shopping_behavior.csv")

df.head()

df.info()

df.describe(include= "all")

df.isnull().sum()

df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))

df.isnull().sum()

df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ','_')
df = df.rename(columns={'purchase_amount_(usd)':'purchase_amount'})


df.columns

#create a column age_group
labels =['young Adults', 'Adult', 'Middle-aged', 'senior']
df['aged_group'] = pd.qcut(df['age'], q=4, labels =labels)

df[['age','aged_group']].head(10)

#create column purchase_frequency_days
frequency_mapping = {
    'fortnightly': 14,
    'weekly': 7,
    'monthly': 30,
    'quaterly': 90,
    'annually': 365,
    'Every 3 months': 90
}
df['purchase_frequency_days'] =
df['frequency_of_purchases'].map(frequency_mapping)

df[['purchase_frequency_days','frequency_of_purchases']].head(10)

df[['discount_applied', 'promo_code_used']].head(10)
```

```
•
• (df['discount_applied'] == df['promo_code_used']).all()
•
• df = df.drop('promo_code_used', axis=1)
• df.columns
```

3. Answering Business Questions with SQL

SQL QUARIES

create database customer_project;

use customer_project;

select * from cleaned_data

limit 20;

#1)what is the total revenue generated by male vs . female customers?

select gender, SUM(purchase_amount) as revenue

from cleaned_data

group by gender

#2) which customers used a discount but still spent more than the average review rating?

select customer_id, purchase_Amount

from cleaned_data

where discount_applied = 'yes' and purchase_amount >= (select avg (purchase_amount) from cleaned_data)

#Q3.which are the top 5 products with the highest average review rating?

SELECT item_purcahsed, ROUND(AVG(review_rating::numeric),2) as "Avereage Product Rating"

from cleaned_data

group by item_purchased

order by avg(review_rating) desc

limit 5;

4)compare teh average purchase amounts between standard and express shipping

select shipping_type,

round(AVG(purchase_amount),2)

from cleaned_data

where shipping_type in ('standard','Express')

group by shipping_type

```
##Q5. Do subscribed customers spend more? Compare average spend and total revenue
##between subscribers and non-subscribers.
select subscription_status,
COUNT(customer_id) as total_customers,
ROUND(AVG(purchase_amount),2) as avg_spend,
ROUND(SUM(purchase_amount),2) as total_revenue
from cleaned_data
group by subscription_status
order by total_revenue, avg_spend desc;
##Q6. Which 5 products have the highest percentage of purchases with discounts applied?
select item_purchased,
ROUND(100 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END)/COUNT(*),2) as
discount_rate
from cleaned_data
group by item_purchased
order by discount_rate desc
limit 5;
###Q7. Segment customers into New, Returning, and Loyal based on their total
##number of previous purchases, and show the count of each segment.
with customer_type as (
select customer_id, previous_purchases,
CASE
   WHEN previous_purchases = 1 THEN 'New'
   WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
   ELSE 'Loyal'
   END AS customer_segment
from cleaned_data
)
select customer_segment, count(*) as "Number of Customers"
from customer_type
group by customer_segment
##Q8. What are the top 3 most purchased products within each category?
```

```
with item_counts as (
select category,
item_purchased,
COUNT(customer_id) as total_orders,
ROW_NUMBER() over(partition by category order by count(customer_id) DESC) as item_rank
from cleaned_data
group by category, item_purchased
)
select item_rank, category, item_purchased, total_orders
from item_counts
where item_rank <= 3;
##Q9. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?
select subscription_status,
count(customer_id) as repeat_buyers
from cleaned_data
where previous_purchases > 5
group by subscription_status
##Q10. What is the revenue contribution of each age group?
select aged_group,
SUM(purchase_amount) as total_revenue
from cleaned_data
group by aged_group
order by total_revenue desc;
select * from cleaned_data;
```

With clean data sitting in MYSQL work Bench, ten specific business questions were answered directly with SQL — aggregations, a correlated subquery, CTEs, and a window function. Each question and its result set is shown below.

1. **Revenue split by gender** — how much total revenue came from male customers versus female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. **Discount users who still spent big** — which customers redeemed a discount but still spent more than the dataset average.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62

Total rows: 839 Query complete 00:00:00.133

3. **Best-rated products** — the five products with the highest average customer review rating.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Standard vs. Express shipping** — whether order value differs between shipping speeds.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. **Subscribers vs. non-subscribers** — how average spend and total revenue compare across subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. **Products most reliant on discounts** — the five products where the largest share of sales involved a discount.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. **Customer loyalty tiers** — grouping customers into New, Returning and Loyal buckets based on purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. **Category best-sellers** — the three top-selling products inside each product category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat buyers and subscriptions** — whether customers with more than 5 past purchases are more likely to be subscribers.

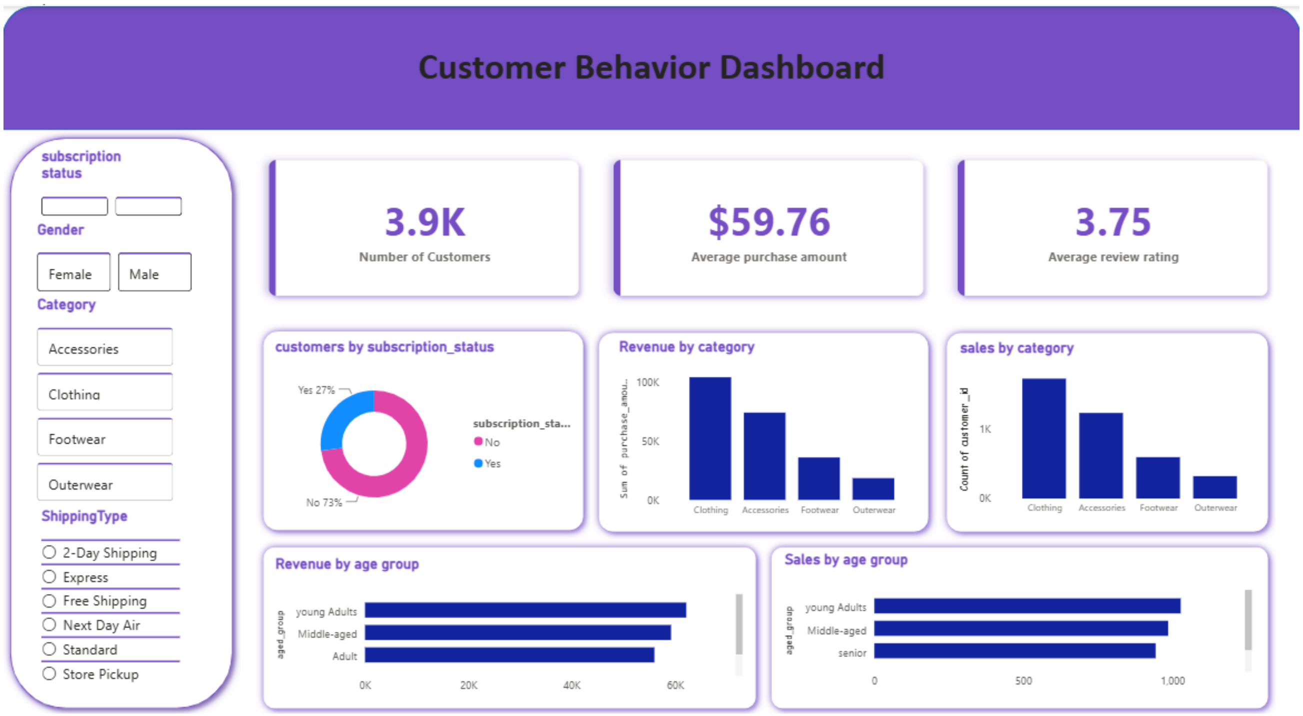
	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. **Revenue by age group** — which age bracket contributes the most total revenue.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

4. Power BI Dashboard

The cleaned dataset was also loaded into Power BI to build a single interactive dashboard, giving anyone on the team a self-service view of the same numbers behind the SQL queries above — filterable by subscription status, gender, category and shipping type.



5. What the Numbers Say

- Male customers accounted for roughly double the revenue of female customers ($157,890 vs. $75,191) — a gap worth digging into further by category and location.
- 839 customers used a discount and still spent above average, so discounting isn't only pulling in bargain-hunters.
- Express shipping orders averaged only about $2 more than Standard ($60.48 vs. $58.46) — shipping speed doesn't meaningfully predict basket size here.
- Subscribers and non-subscribers spend almost the same per order (~$59-60); non-subscribers only generate more total revenue because there are nearly three times as many of them.
- 80% of the customer base is already “Loyal” (10+ past purchases), against just 2% brand-new customers — retention is strong, acquisition looks like the growth lever.
- Pants, Blouse and Jewelry lead their categories in order volume; Gloves, Sandals and Boots lead in customer satisfaction.

6. Recommendations

- **Rework the subscriber value proposition —** the current subscription program isn't lifting average spend, so it may be worth adding a clearer incentive (e.g. free shipping, early access) rather than just the badge of being subscribed.
- **Invest more in new-customer acquisition —** 98% of customers are repeat or loyal, which is great for retention but signals the funnel isn't bringing in new customers — worth a dedicated acquisition push.
- **Audit discount depth on high-discount-rate products —** since near-50% discount rates on items like Hats and Sneakers may be eating into margin without necessarily driving incremental volume.
- **Lean on top-rated products in marketing —** use Gloves, Sandals and Boots (highest-rated) in campaigns and cross-sell them alongside lower-rated items in the same category.

- **Look deeper into the gender revenue gap —** the ~$80K gap between male and female revenue is large enough to warrant a follow-up analysis by category before deciding on any campaign.